UNIVERSITAT POLITÈCNICA DE CATALUNYA

9.5

E.T.S. d'Enginyeria de Telecomunicació de Barcelona

E.T.S. d'Enginyers de Camins, Canals i Ports de Barcelona

Facultat d'Informàtica de Barcelona

Titulació

Assignatura

Cognoms: Cullell Martínez   Nom: Arnau

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DNI

CCQ: EXERCICI 1

cap càlcul explícit polar → cartesià −0.5

1) Donats els nombres complexos $\bar{z}_1 = 5+3i$, $\bar{z}_2 = -2-5i$ calcular i expressar en forma cartesiana i polar:

a) $\bar{z}_1 + \bar{z}_2 = (5-2)+(3-5)i = \boxed{3-2i}$; $r = \sqrt{3^2+(-2)^2} = \sqrt{13}$, $\theta = tg^{-1}\left(\frac{-2}{3}\right) = -33,7$; $\boxed{\sqrt{13}\,\lfloor -33,7}$ ✓

b) $\bar{z}_1 - \bar{z}_2 = (5-(-2))+(3-(-5)) = \boxed{7+8i}$; $r = \sqrt{7^2+8^2} = \sqrt{113}$, $\theta = tg^{-1}\left(\frac{8}{7}\right) = 48,81$; $\boxed{\sqrt{113}\,\lfloor 48,81}$ ✓

c) $\bar{z}_1 \times \bar{z}_2$; $\bar{z}_1$ polar expo $= \sqrt{34}\,\lfloor 30,96$; $\bar{z}_2$ polar expo $= \sqrt{29}\,\lfloor 68,20$ → $68,2-180 = -111,8$

$\bar{z}_1 \cdot \bar{z}_2 = \sqrt{34}\cdot\sqrt{29}\,\lfloor 30,96+(-111,8) = \boxed{\sqrt{986}\,\lfloor -80,84}$?$= \boxed{5-31i}$

d) $\frac{\bar{z}_1}{\bar{z}_2} = \frac{\sqrt{34}}{\sqrt{29}}\,\lfloor 30,96-(-111,8) = \boxed{\sqrt{\frac{34}{29}}\,\lfloor 142,76} = \boxed{-0,862+0,655i}$ ✓

e) $\bar{z}_1 + \bar{z}_2^* = (5-2)+(3+5) = \boxed{3+8i}$; $r = \sqrt{3^2+8^2} = \sqrt{73}$, $\theta = tg^{-1}\left(\frac{8}{3}\right) = 69,44$; $\boxed{\sqrt{73}\,\lfloor 69,44}$ ✓

f) $\frac{\bar{z}_1}{\bar{z}_2^*} = \frac{\sqrt{34}}{\sqrt{29}}\,\lfloor 30,96-111,8 = \boxed{\sqrt{\frac{34}{29}}\,\lfloor -80,84} = \boxed{0,172-1,068i}$ ✓

2) Considerem la funció complexa $\bar{f}(x,t) = Ae^{ik(x-vt)}$, calculem $\frac{\partial \bar{f}}{\partial x}$, $\frac{\partial^2 \bar{f}}{\partial x^2}$, $\frac{\partial f}{\partial t}$, $\frac{\partial^2 f}{\partial t^2}$ i comprovem si es verifica o no l'equació d'ones $\frac{\partial^2 \bar{f}}{\partial x^2} - \frac{1}{v^2}\cdot\frac{\partial^2 \bar{f}}{\partial t^2} = 0$

- $\frac{\partial \bar{f}}{\partial x} = \boxed{Ae^{ik(x-vt)}\cdot ik}$

- $\frac{\partial^2 \bar{f}}{\partial x^2} = Ae^{ik(x-vt)}\cdot ik\cdot ik = Ae^{ik(x-vt)}\cdot i^2k^2 \underset{i^2=-1}{=} \boxed{Ae^{ik(x-vt)}\cdot -k^2}$

- $\frac{\partial f}{\partial t} = \boxed{A e^{ik(x-vt)} \cdot -ikv}$

- $\frac{\partial^2 f}{\partial t^2} = A e^{ik(x-vt)} \cdot + i^2 k^2 v^2 \underset{i^2=-1}{=} \boxed{A e^{ik(x-vt)} \cdot -k^2 v^2}$

- $\frac{\partial^2 f}{\partial x^2} - \frac{1}{v^2}\frac{\partial^2 f}{\partial t^2} \equiv \frac{\partial^2 f}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 f}{\partial t^2} \underset{\text{Substituim}}{\equiv} A e^{ik(x-vt)} \cdot -k^2 = \frac{1}{\cancel{v^2}} \cdot A e^{ik(x-vt)} - k^2 \cdot \cancel{v^2}$

$\equiv A e^{ik(x-vt)} \cdot (-k^2) = A e^{ik(x-vt)} \cdot (-k^2)$ es compleix!

3) Feu les operacions i expreseu en forma cartesiana:

1. $e^{-i\pi} \cdot 5 \cdot e^{-i\frac{\pi}{4}} = 5 \cdot e^{(-i\pi - i\frac{\pi}{4})} = 5 \cdot e^{-i(\pi + \frac{\pi}{4})}$

Forma cartesiana: $5\left(\cos\left(\pi+\frac{\pi}{4}\right) + i \sin\left(\pi+\frac{\pi}{4}\right)\right) = \boxed{5\left(-\frac{\sqrt{2}}{2}\right) + 5\left(\frac{\sqrt{2}}{2}\right) i}$

2. $e^{-i\frac{\pi}{2}} \cdot \left(2e^{i\frac{\pi}{2}} + 3e^{-i\frac{\pi}{2}}\right) \underset{\substack{x = r\cos\theta \\ y = r\sin\theta}}{=} e^{-i\frac{\pi}{2}} \cdot (0 + 2i + 0 - 3i) = e^{-i\frac{\pi}{2}} \cdot (-i) \underset{\substack{x = r\cos\theta \\ y = r\sin\theta}}{=} (-i)\cdot(-i) \underset{i^2=-1}{=} \boxed{-1}$

3. $\frac{1}{e^{i\frac{3\pi}{2}}} \cdot \left(4e^{-i\frac{\pi}{2}} - 2e^{i\pi}\right) \underset{\substack{x = r\cos\theta \\ y = r\sin\theta}}{=} \frac{1}{e^{i\frac{3\pi}{2}}} \cdot \left((0 - 4i) - (-2 + 0i)\right) = e^{-i\frac{3\pi}{2}} \cdot (+2 - 4i) \underset{\substack{x = r\cos\theta \\ y = r\sin\theta}}{=} i(2 - 4i)$

$\underset{i^2=-1}{=} 2i - 4i^2 = \boxed{4 + 2i}$

4) Fent ús del desenvolupament en sèrie de Taylor d'una funció $f(x)$, per a punts propers a $x=0$:
$$f(x) = f(0) + x \cdot \frac{df(x)}{dx}\bigg|_{x=0} + \frac{1}{2!} x^2 \cdot \frac{d^2 f(x)}{dx^2}\bigg|_{x=0} + \ldots$$
Calculeu el desenvolupament prop de $x=0$ per a $\sin(x)$ i $\cos(x)$:

- $\sin(x)$: $0 + x \cdot \cos(0) + \frac{1}{2!} x^2 \cdot -\sin(0) + \frac{1}{3!} x^3 \cdot -\cos(0) + \frac{1}{4!} x^4 \cdot \sin(0) + \ldots$

Sabent que $\sin(0) = 0$ i que $\lim_{x\to 0} \sin(x) = \sin(0) = 0$ obtenim que:
$\cos(0) = 1$

$$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots$$

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- $\cos(x) = 1 + x \cdot -\sin(x) + \frac{1}{2!} x^2 \cdot -\cos(x) + \frac{1}{3!} x^3 \cdot \sin(x) + \frac{1}{4!} x^4 \cdot \cos(x) + \dots$

Anàlogament com abans:

$$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

5) Calculeu el desenvolupament en sèrie de Taylor de la funció e^{ix} prop de $x=0$ i comproveu que es verifica $e^{ix} = \cos(x) + i\sin(x)$:

- $e^{ix} = 1 + x \cdot e^{ix} \cdot i + \frac{1}{2!} \cdot x^2 \cdot e^{ix} \cdot (i)^2 + \frac{1}{3!} x^3 \cdot e^{ix} \cdot (i)^2 \cdot (i) + \frac{1}{4!} x^4 \cdot e^{ix} \cdot (i)^2 \cdot (i)^2 + \dots$

Sabent que $i^2 = -1$ i que $\lim_{x \to 0} e^{ix} = e^{i \cdot 0} = 1$ tenim:

$$1 + xi - \frac{x^2}{2!} - \frac{x^3}{3!} i + \frac{x^4}{4!} + \dots$$

$$e^{ix} = 1 + xi - \frac{x^2}{2!} - \frac{x^3}{3!} i + \frac{x^4}{4!} + \frac{x^5}{5!} i - \frac{x^6}{6!} + \dots$$

$$e^{ix} = \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots\right) + i\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots\right)$$

$$\boxed{e^{ix} = \cos(x) + i \cdot \sin(x)}$$